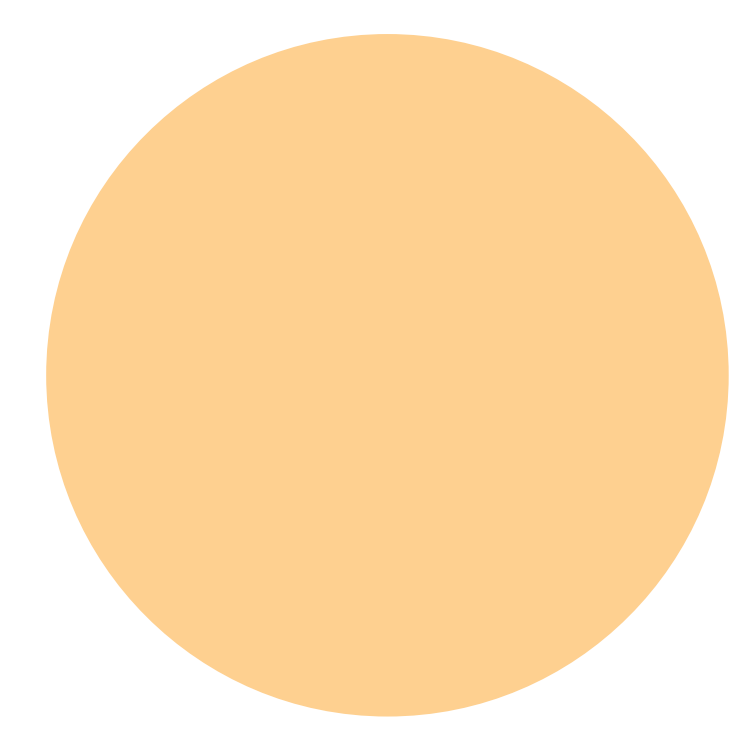




the hatfield wa[y]ste

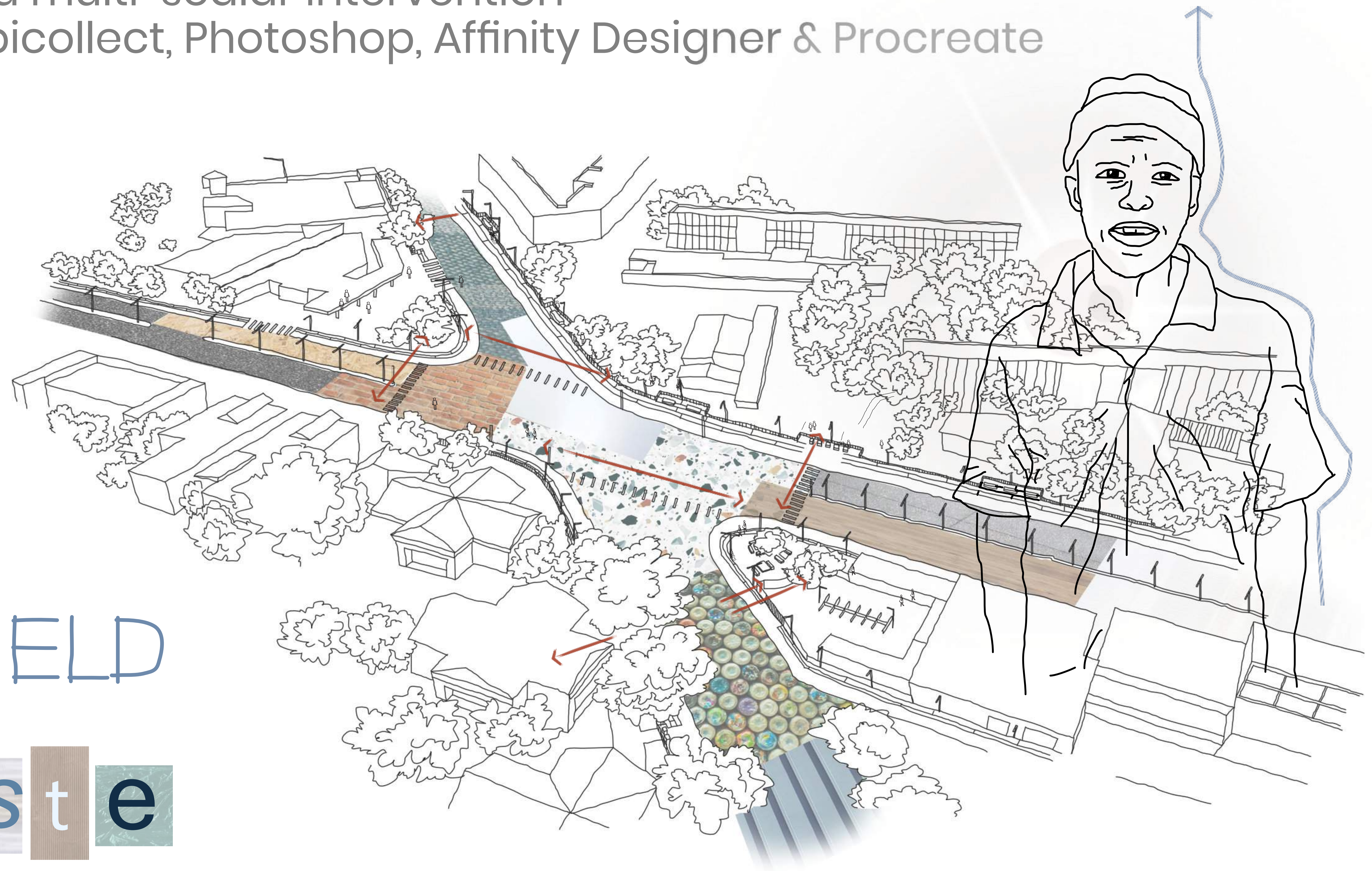
future cities systems

Project Timeline: Q1 Honours Project [2023]

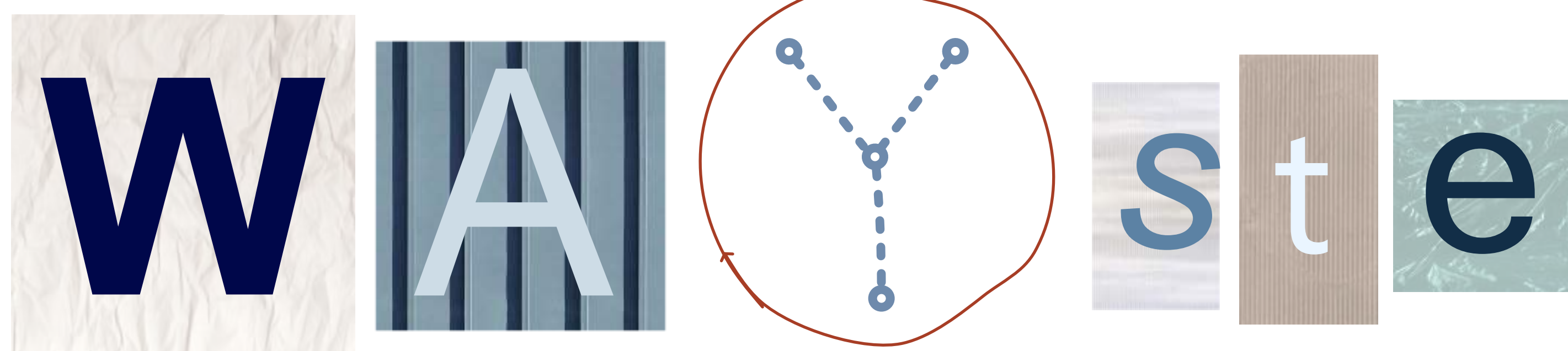
Location: Hatfield, South Africa

Approach: Phased multi-scalar intervention

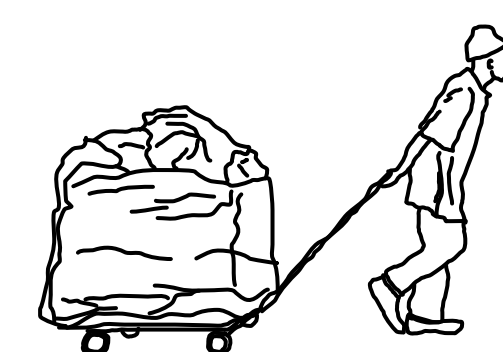
Software: QGIS, Epicollect, Photoshop, Affinity Designer & Procreate



THE HATFIELD



A BAGAREZI'S REIMAGINED WORLD



Project Overview

the brief

Developing a *visionary spatial strategy* for the greater Hatfield area in Pretoria, using *food systems as a driver* for *sustainable urban development*. By analysing current networks, the project seeks *systemic, catalytic solutions* that address *density, localisation, circular economy, technology, and culture to shape healthier, more resilient African cities*.

the problem identified

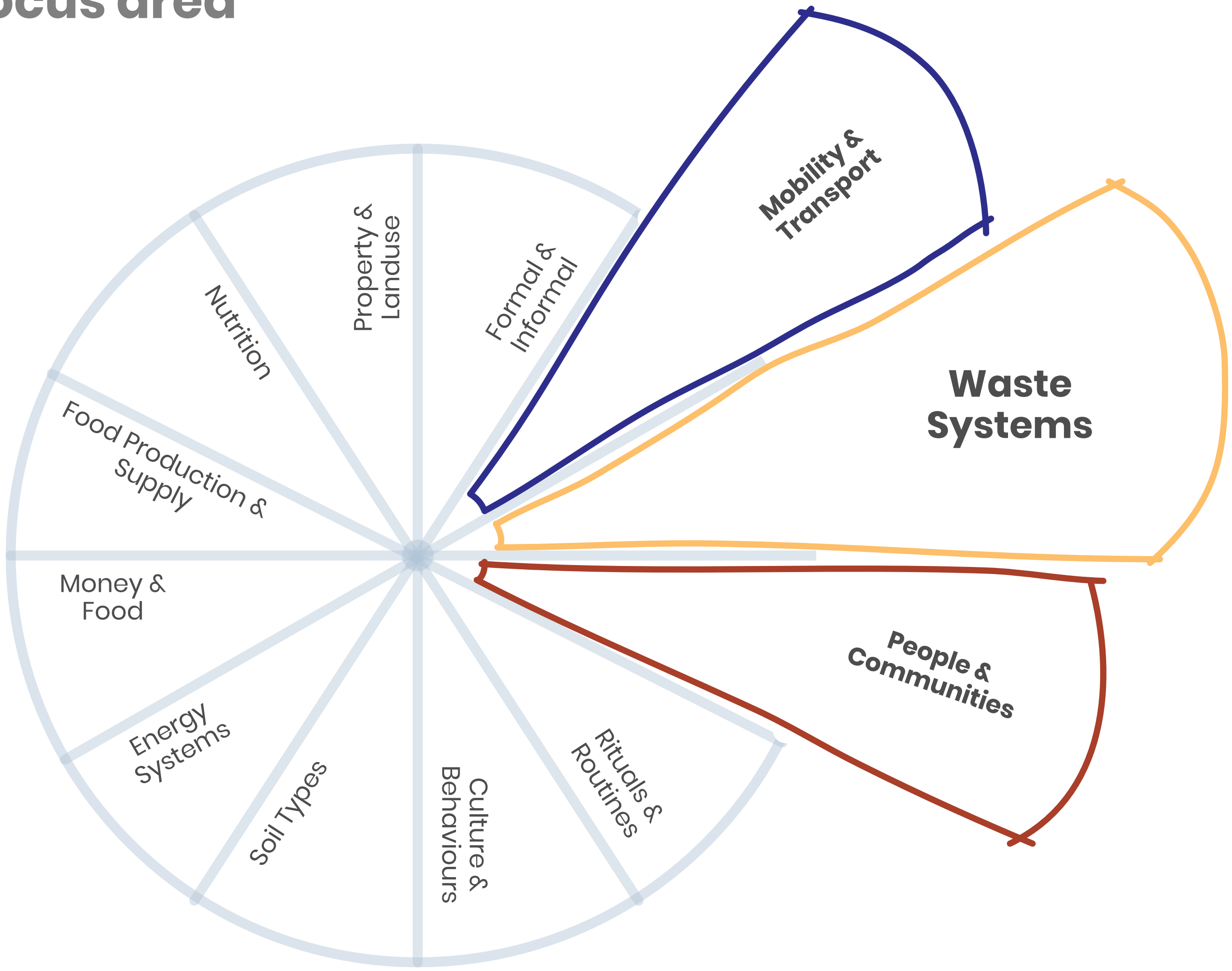
The project’s problem is twofold:
Firstly, Hatfield’s current waste system operates as a *linear trash economy* where food enters from large distribution centres and waste exits to municipal landfills or formal recyclers, leaving the informal waste-picking economy – the Bagarezi – excluded from its benefits and structures.

Secondly, the Bagarezi face *severe home and work insecurity*. Within the Hatfield precinct, there are no designated informal sorting facilities, only scattered waste piles. Without recognition, infrastructure, or affordable, legal spaces to work and live, the Bagarezi remain *marginalised despite playing a crucial role in recycling*.

the solution

Achieving recognition for the Bagarezi and integrating them into a collaborative trash economy through a *systemic spatial strategy* in the Hatfield precinct – both benefiting the precinct and the informal waste economy.

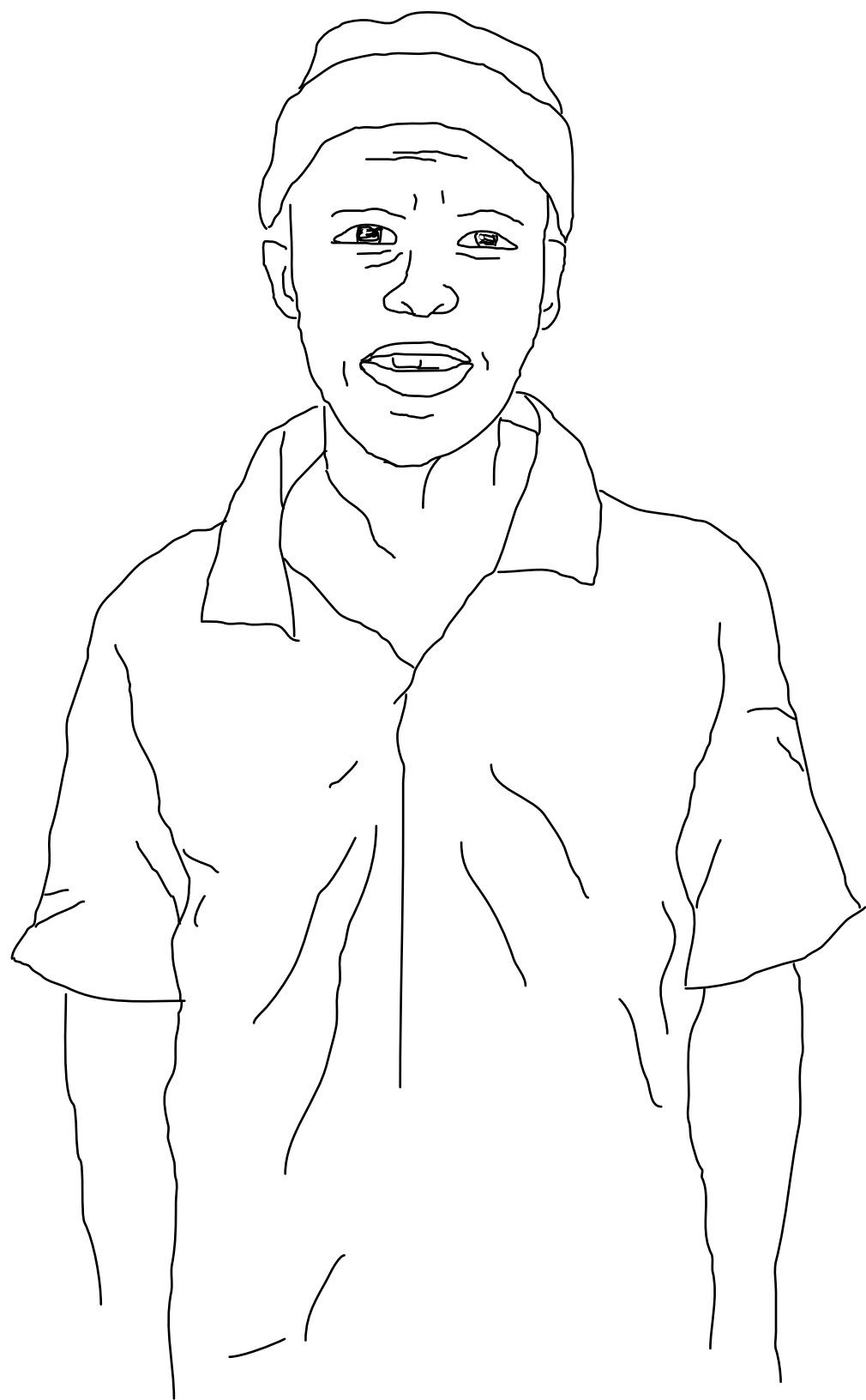
food system focus area



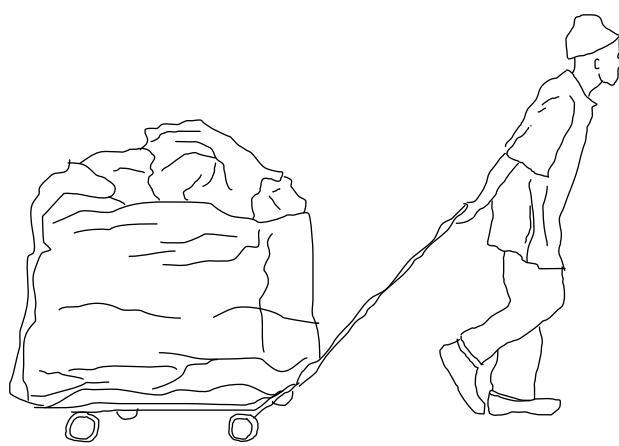
the Bagarezi

Who are the Bagarezi’s?

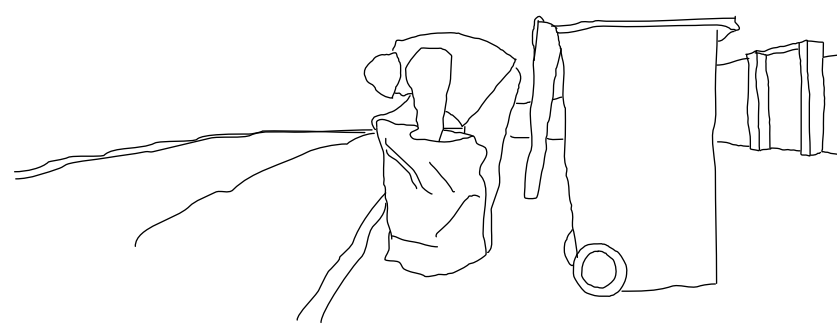
The Bagarezi [translating to independent, self-driven and innovative workers in Shona] make a living by collecting recyclable materials from suburban rubbish bins and selling them on to middlemen and big recyclers.



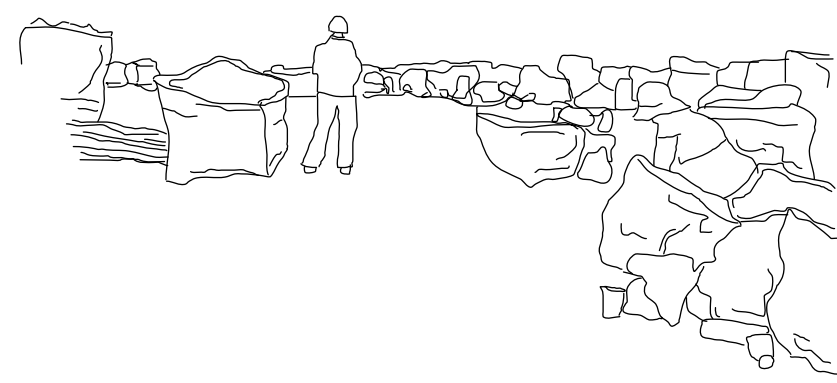
Existing process



Traveling



Collecting



Sorting



Weighing & inspecting

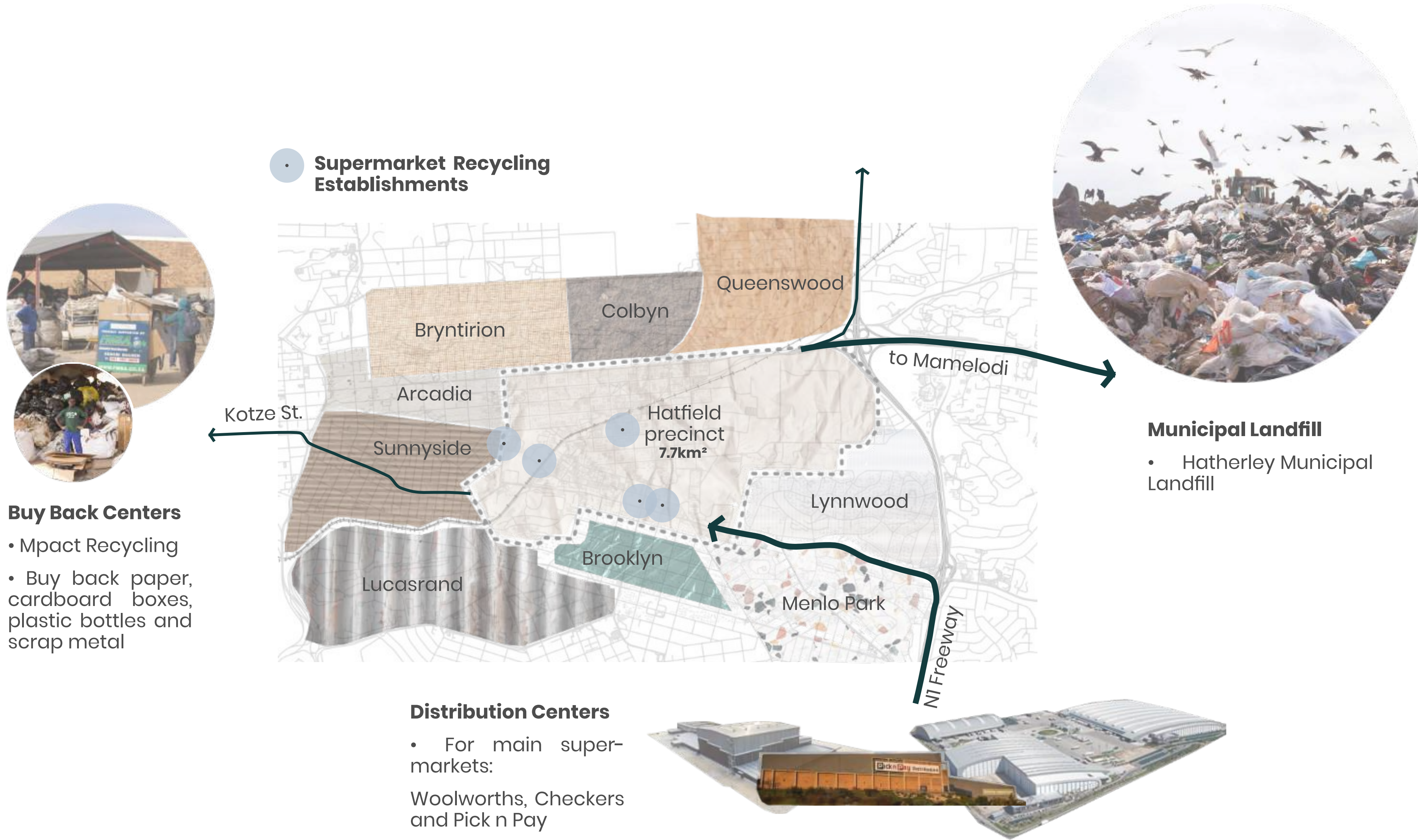


Exchanging

Site Study

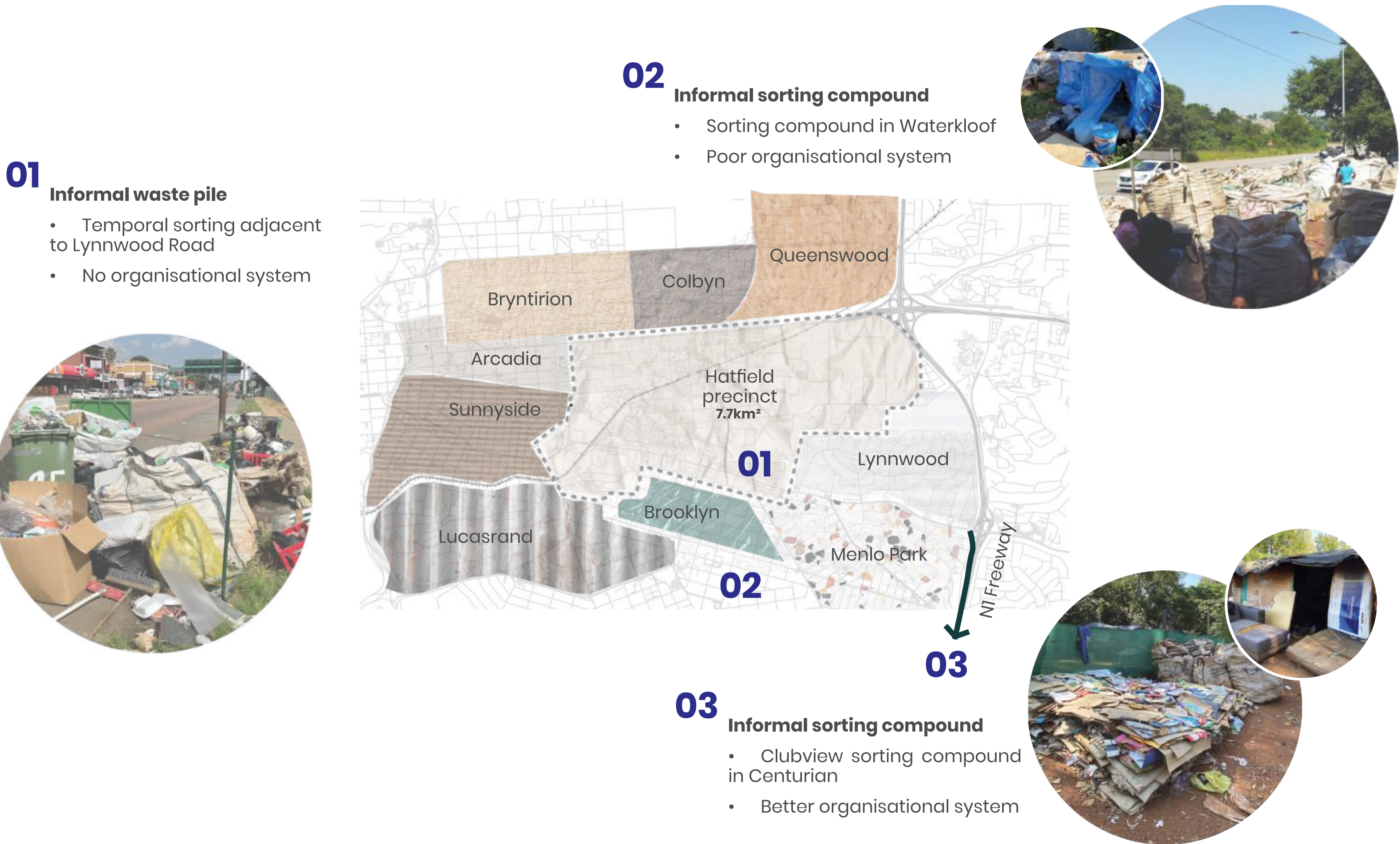
linear trash economy problem

How can we include the informal trash economy into this system?



home & work insecurity problem

How can we create established informal sorting compounds, better infrastructure and homes for the Bagarezi within the precinct?



The Hypothesis

If we can achieve recognition for the Bagarezi and integrate them into a collaborative trash economy, the waste management system will improve and livelihoods will be uplifted.

mapping focus area

Hatfield Precinct, Pretoria, South Africa.



- | | |
|---------------------------------------|----------------------------|
| Legend | Tshwane Legend |
| • Garages | ■ Buildings |
| ■ 5 min walking distance from Garages | — Roads |
| ■ Freestanding residential unit zones | — Railway |
| — Pedestrian pathways | — Main highway |
| ■ Vacant land | --- Precinct boundary line |

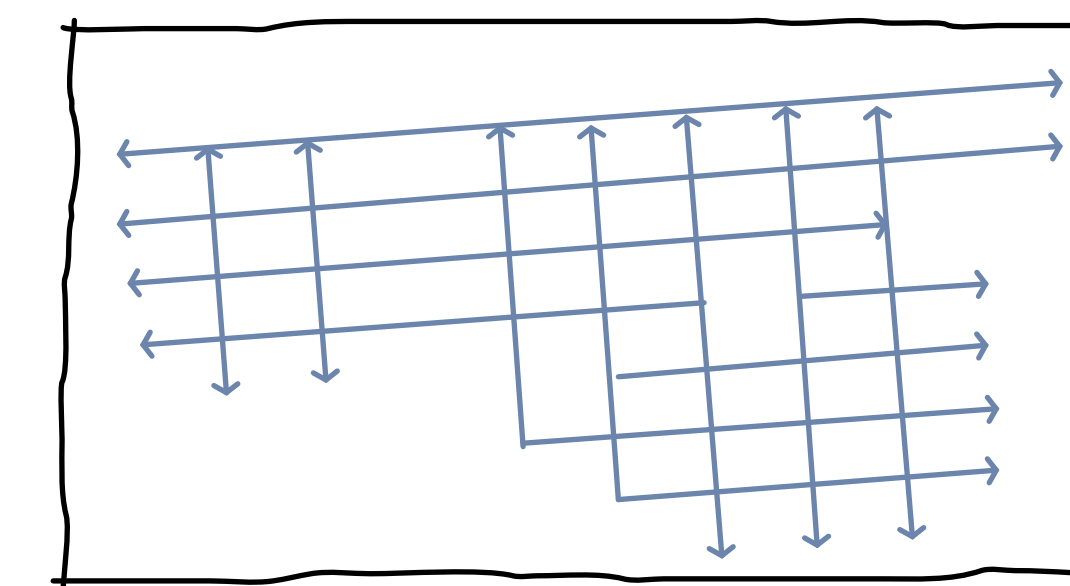
The site investigation focuses on the *free-standing residential units and garages* which contribute to the waste collection of the Bagarezis. *Pedestrian pathways* were looked at in order to determine which routes waste-pickers can mainly access. *Vacant land* was included as a means for opportunity to occur.

The site investigation makes us raise questions such as:

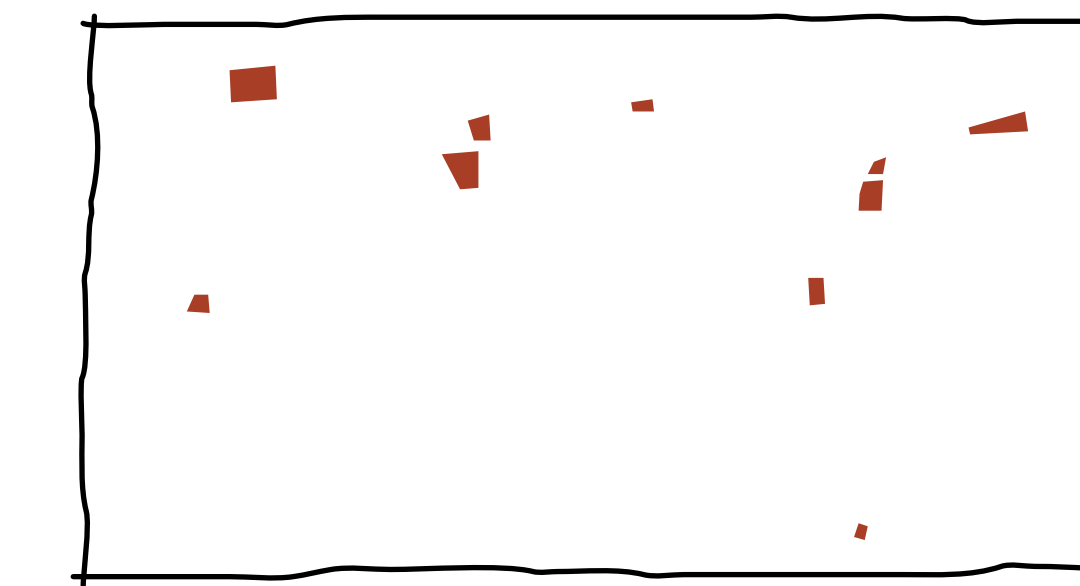
- How can we transform vacant land into meaningful land?
- Connect similar zones and facilities?
- Link disconnected pathways?
- Make use of large open land space in a way that creates a collaborative trash economy within the precinct?

analysis

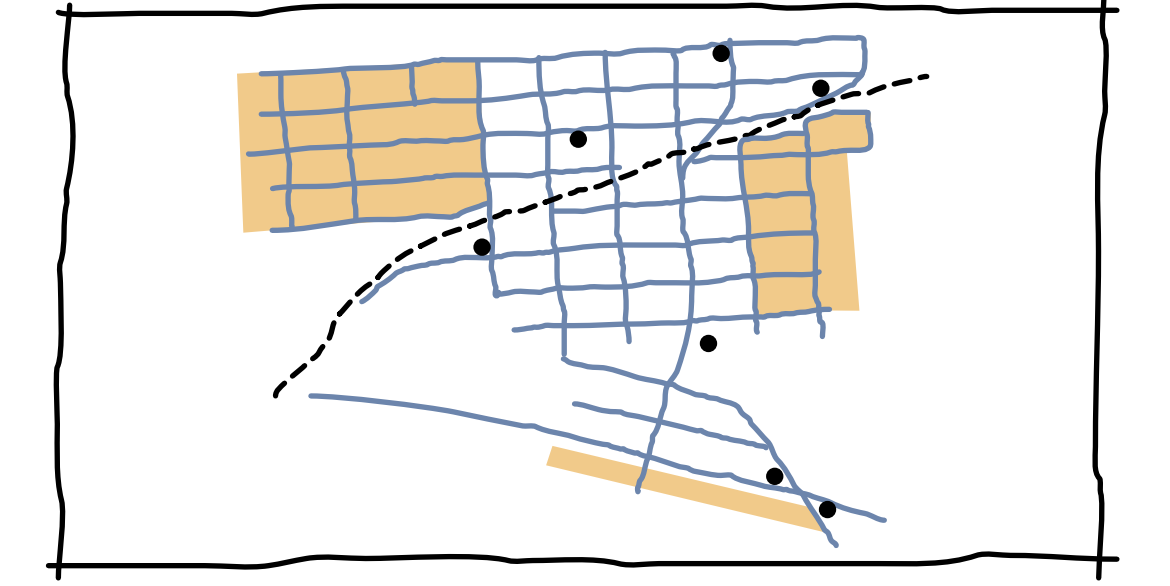
Diagrammatic analysis



Restricted linear movement.

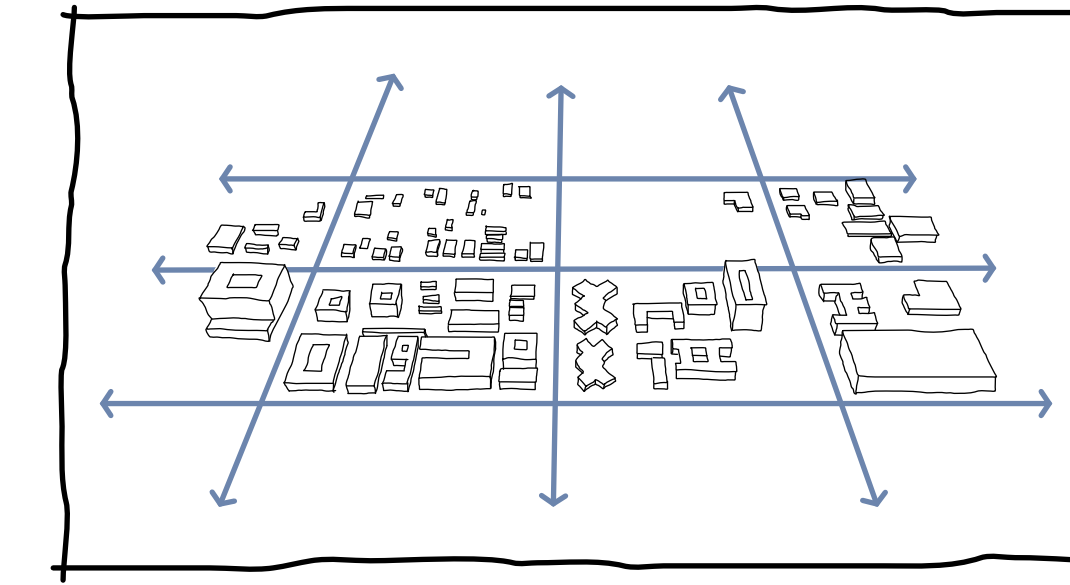


Isolated vacant land.

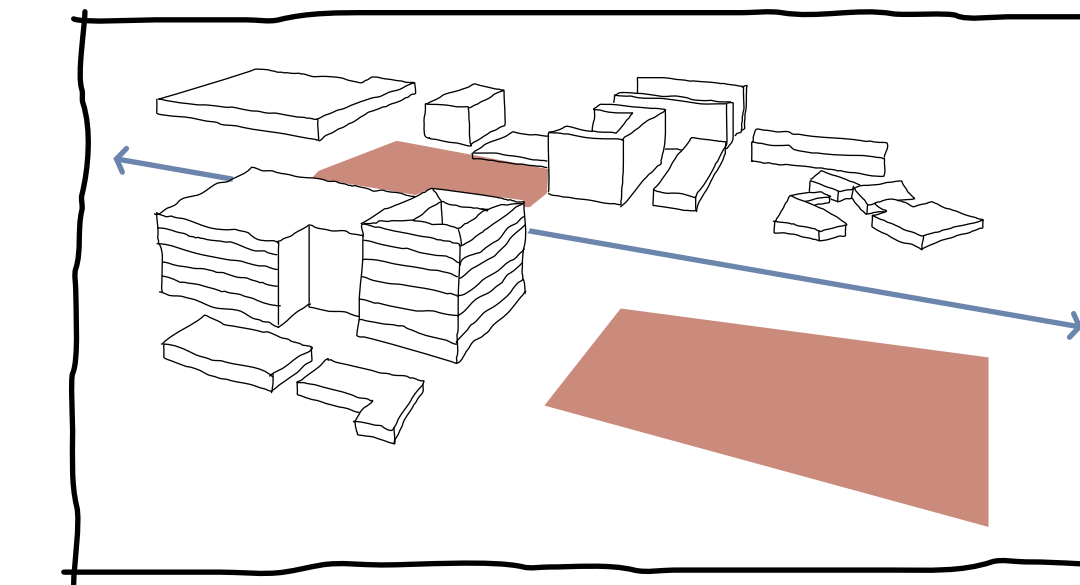


Disconnected facilities and zoning

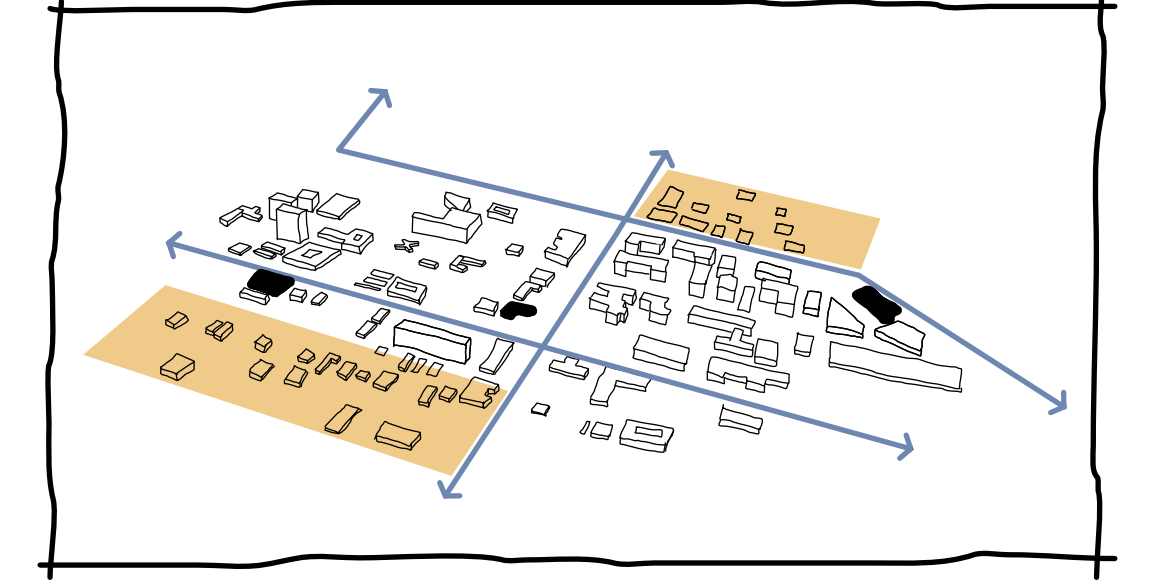
Volumetric analysis



Restriction to orthogonal layout hindering direct routes from one point to another.

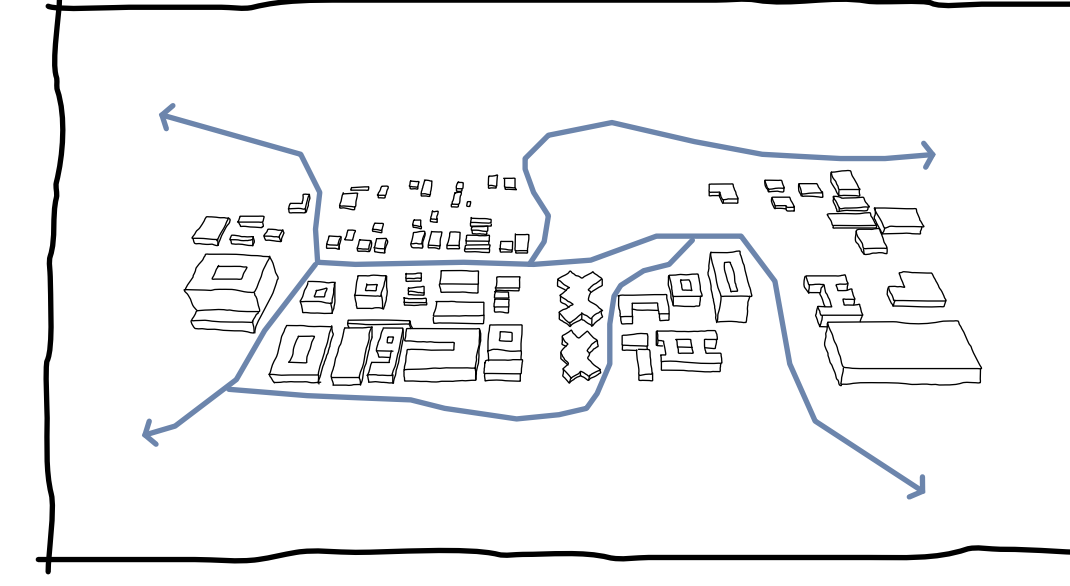


Vacant land between pathways creates dead zones.

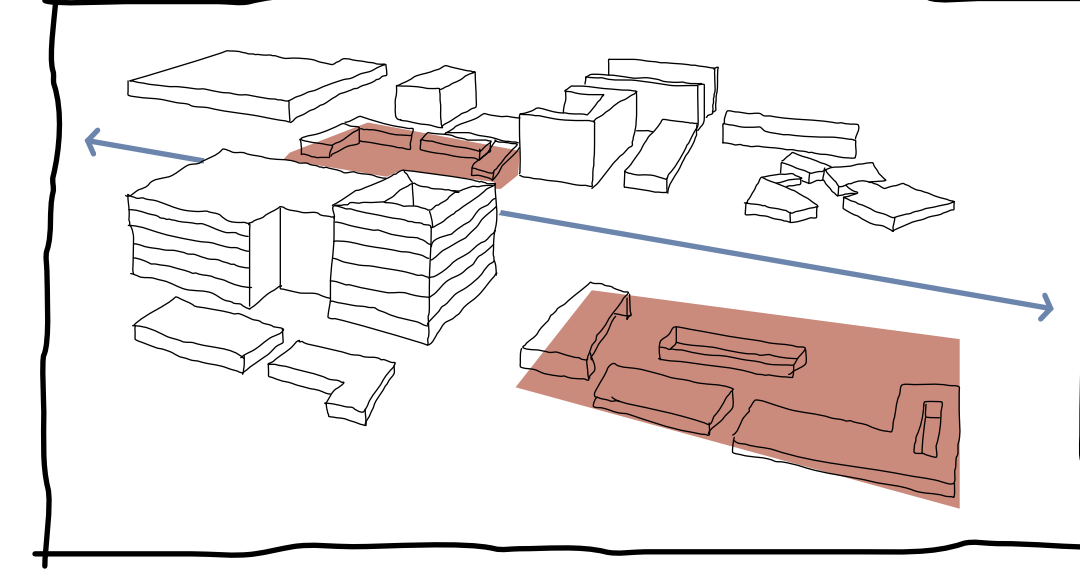


No clear connection between similar facilities and zoning.

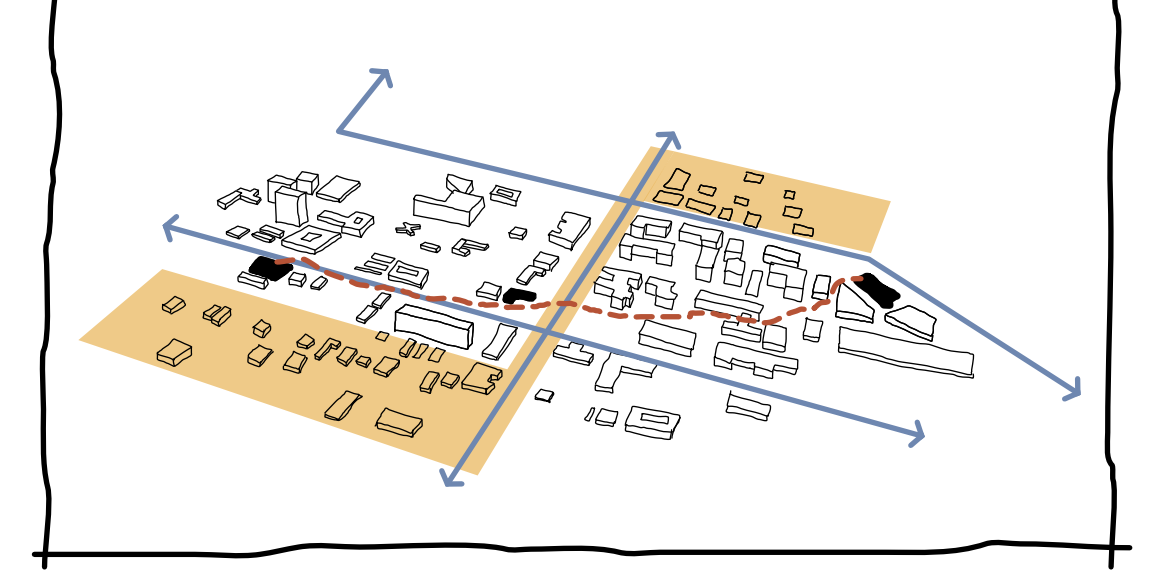
Interpreting literature



- "Avoid long pedestrian routes" (Gehl, 2011:141)
- "The paths are given identity through the edges they move along" (Lynch, 1960:41).

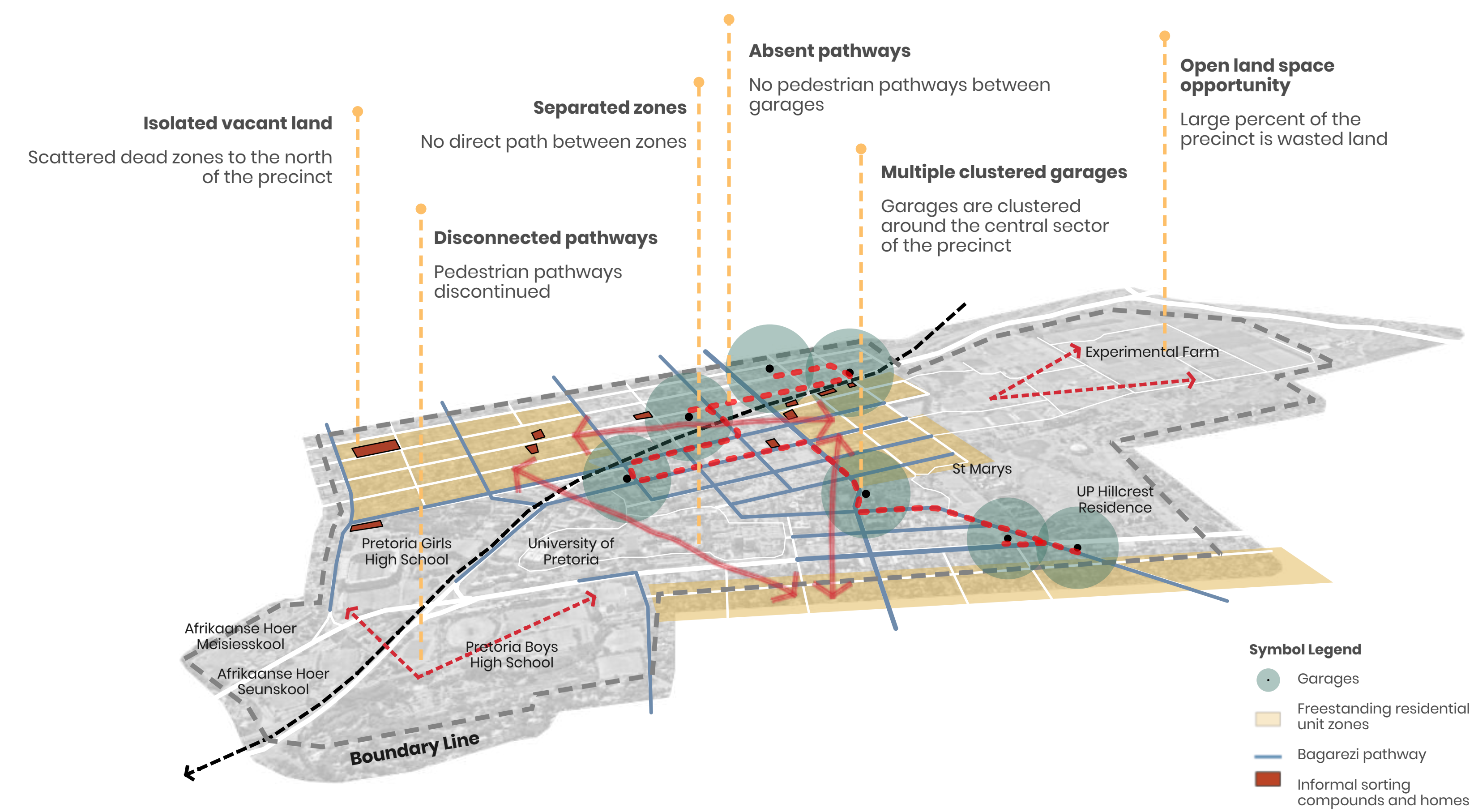


- "Living cities are always stimulating because they are rich in experiences" (Gehl, 2011:21).

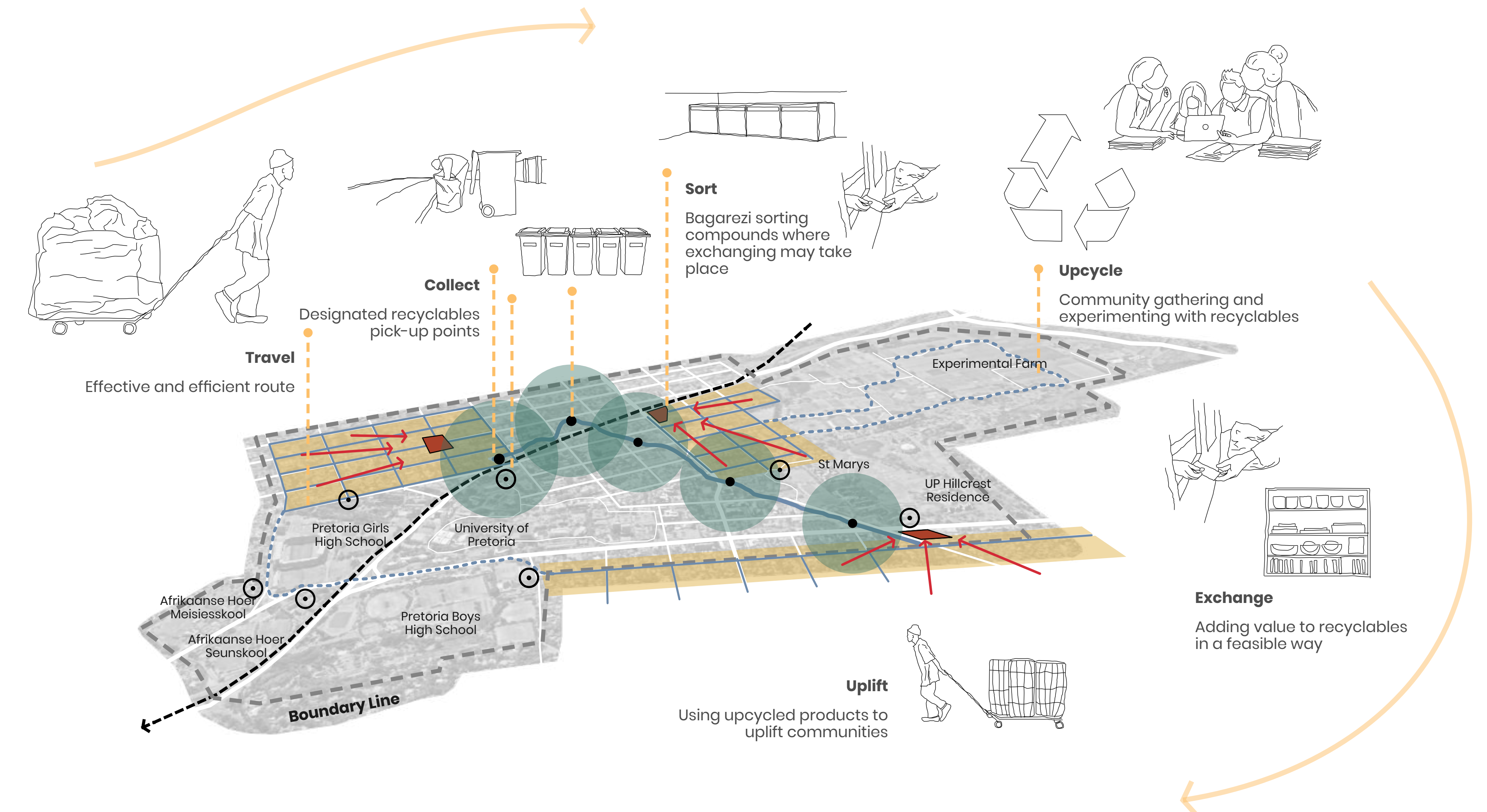


- "Break building blocks in manageable stages" (Gehl, 2011:141)
- "Edges and paths call for certain continuity" (Lynch, 1960:41).

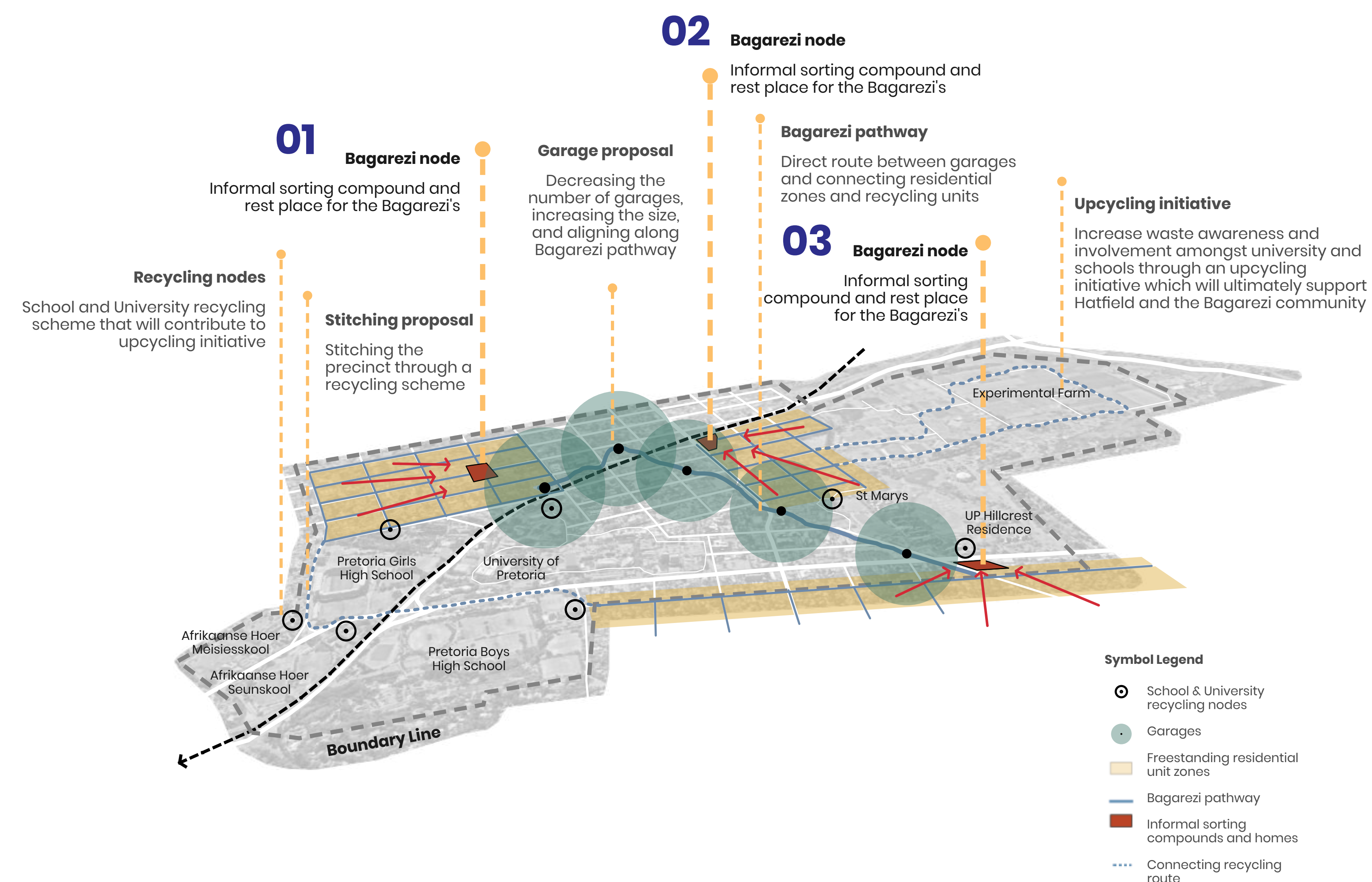
opportunities



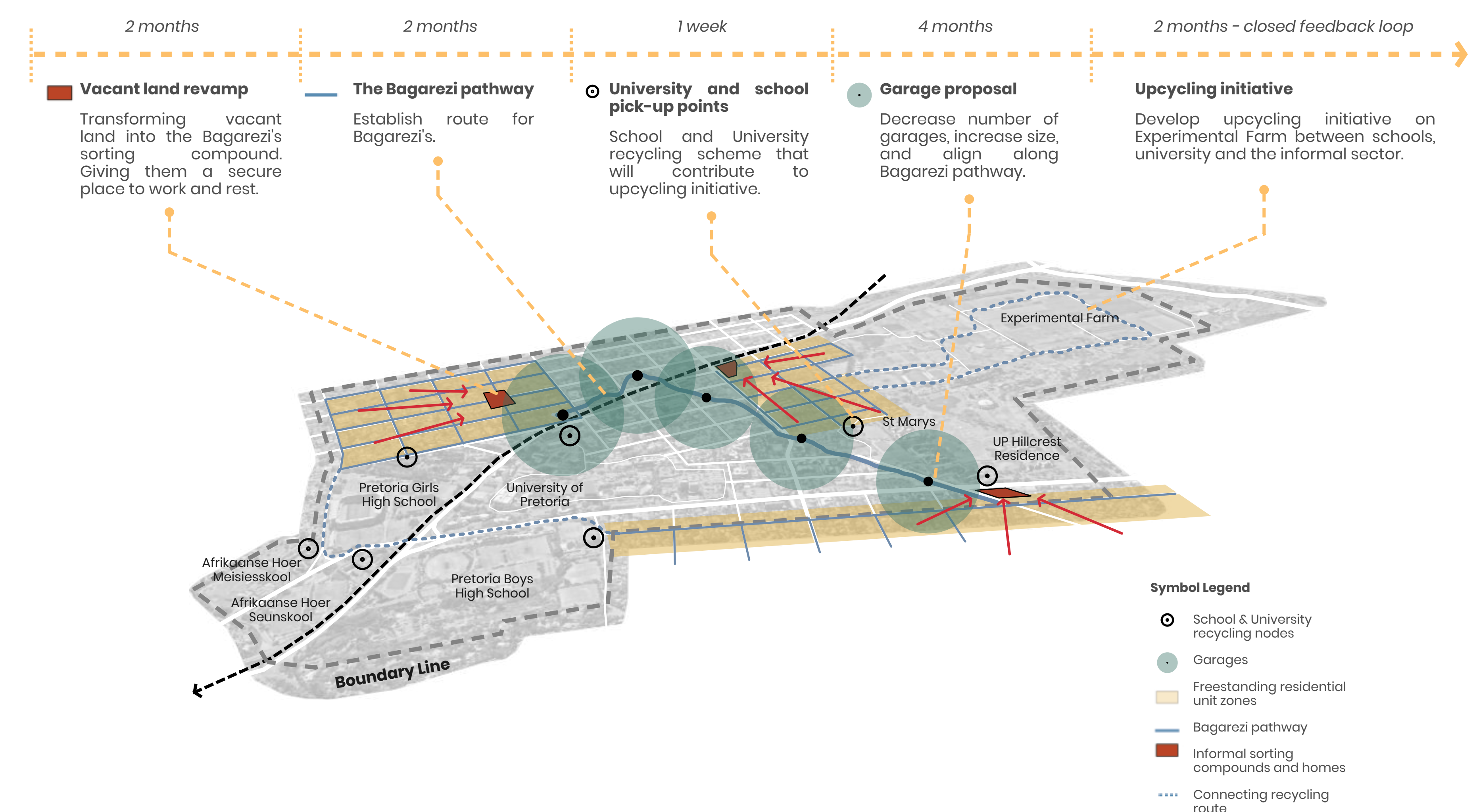
collaborative trash economy



site intervention



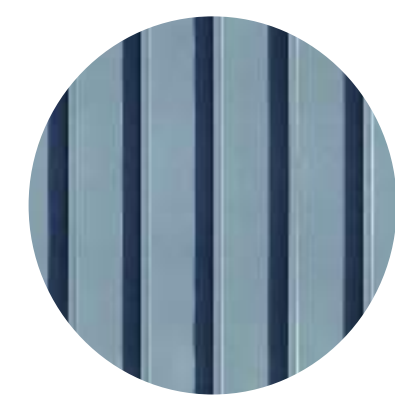
phased timeline



Design Sets

materiality

Materials that can be or have been recycled



Metal

Recycled truck containers



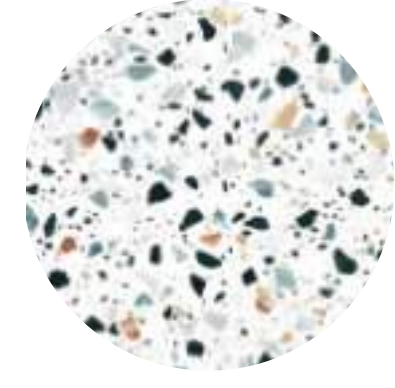
Black powder-coated steel



Plastic



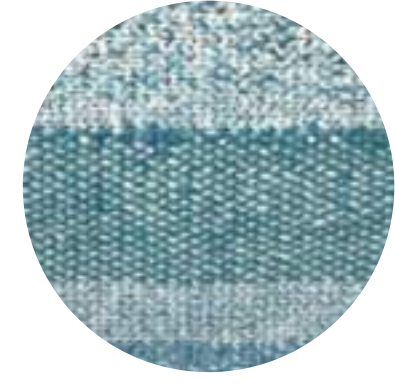
Eco-bricks



Plastic terrazzo



Asphalt



Recycled plastic bags



Timber

Reclaimed timber



Oriented strand board [OSB]



Glass

Glass



Stone & Ceramic

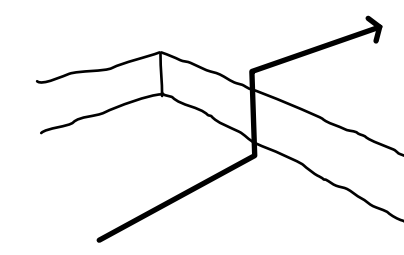
Unfired clay brick



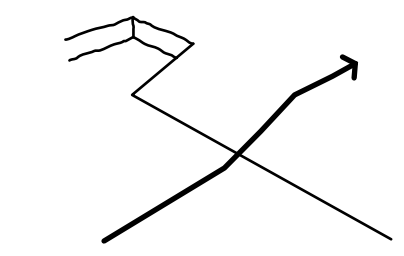
Stone aggregate tile

street parts

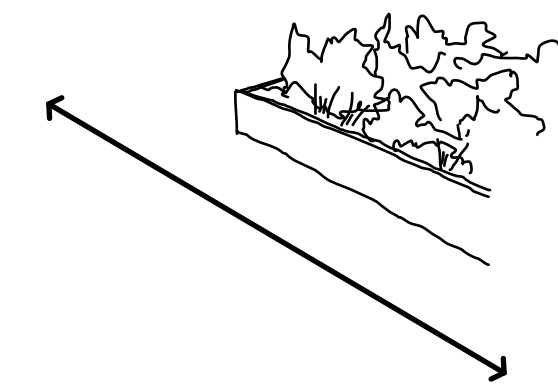
Floor levels



Stepped kerb

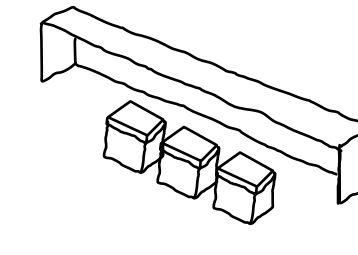


Ramped kerb

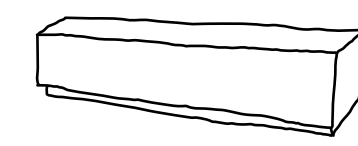


Flat surface

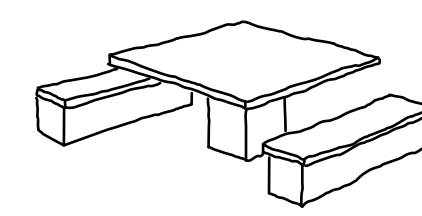
Table & seating



Individual seating with table space

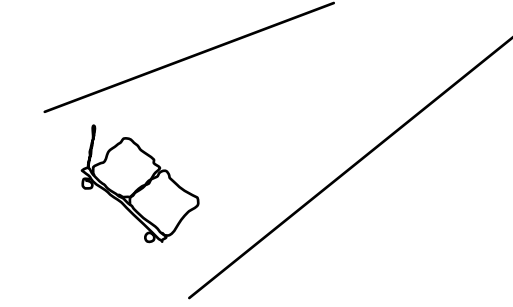


Low bench

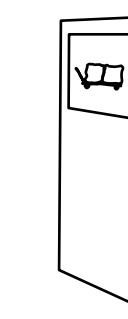


Communal seating

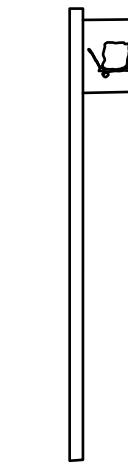
Signage



Floor surface

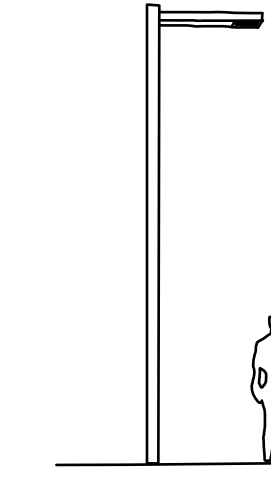


Eye-level

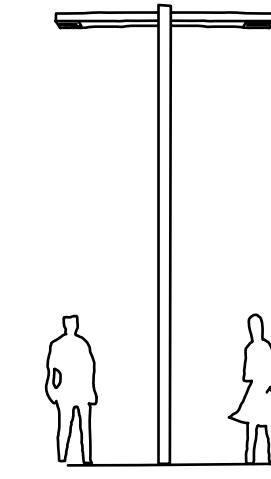


Distant

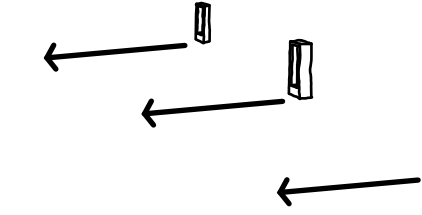
Light



Single overhead

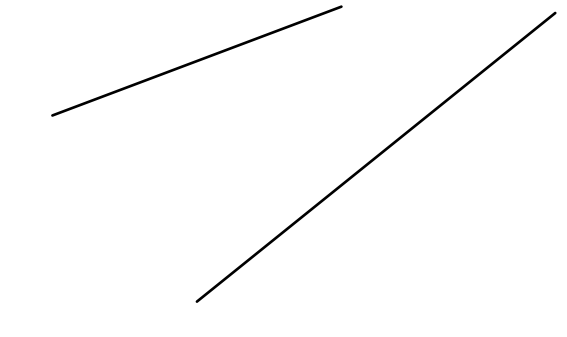


Double overhead

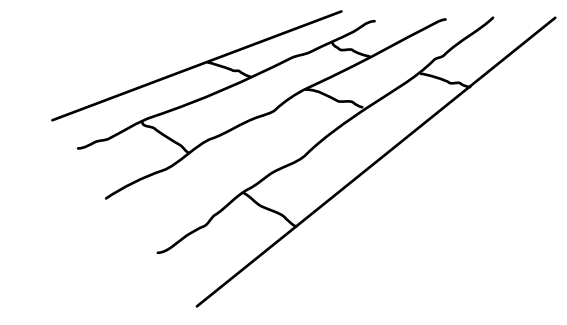


Floor level

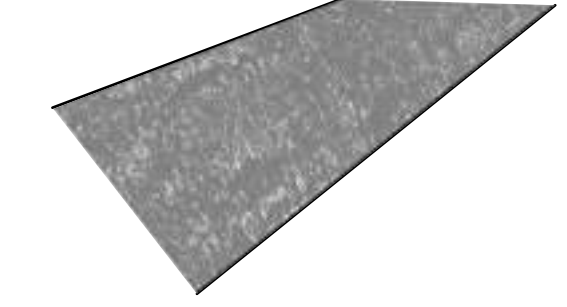
Floor tactility



Smooth



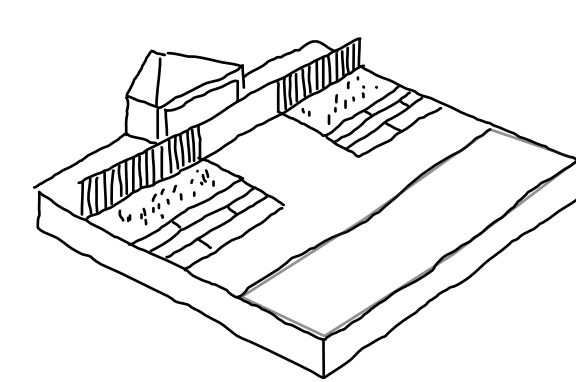
Tiled



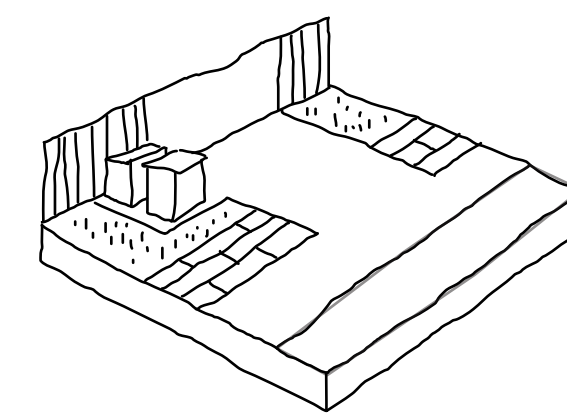
Grainy

processes

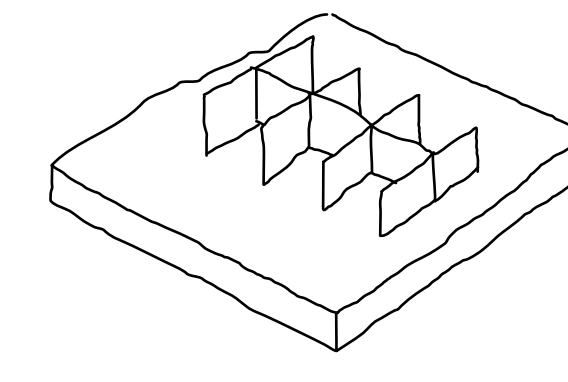
Travel



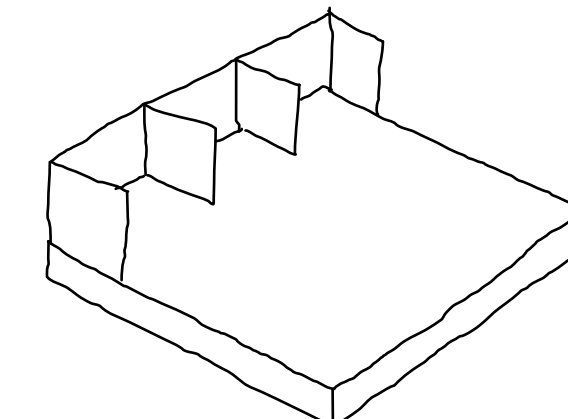
Collect



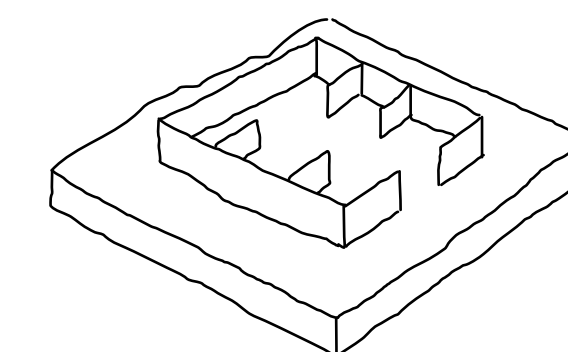
Sort



Middle

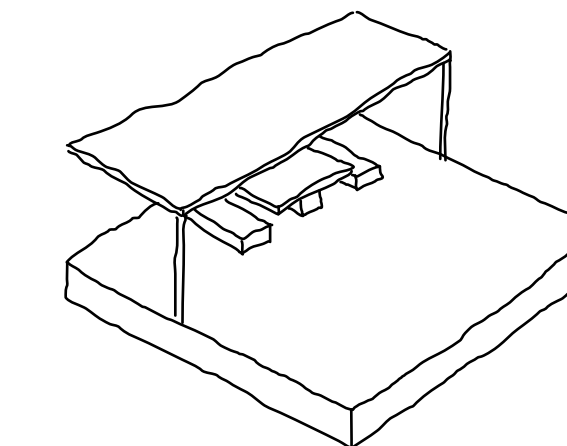


Outskirts

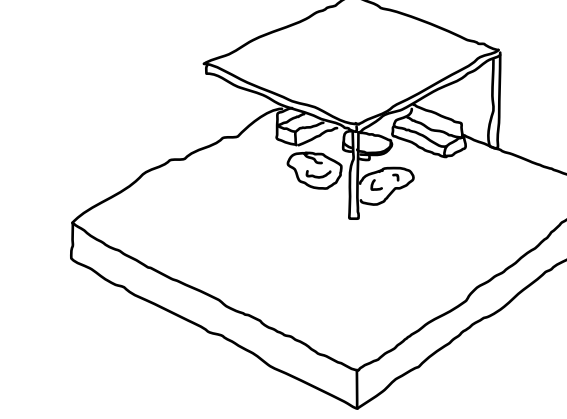


Courtyard

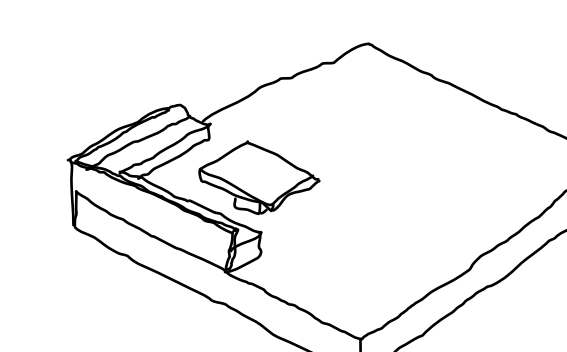
Gather



Conference & workshop

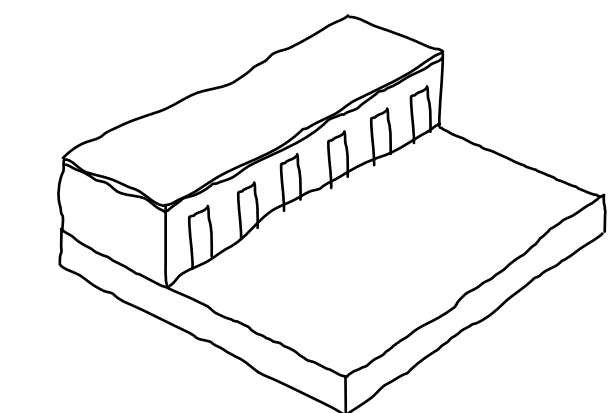


Chill

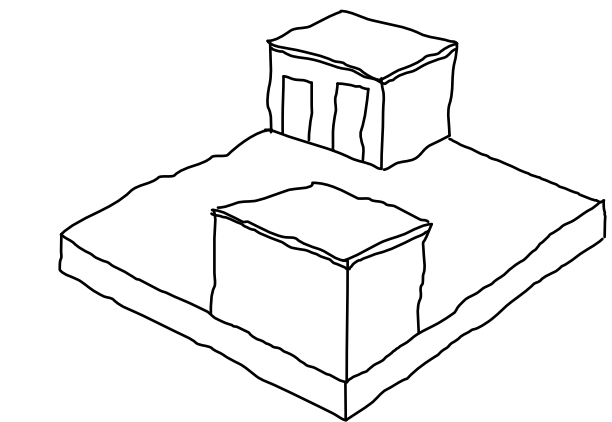


Socialize

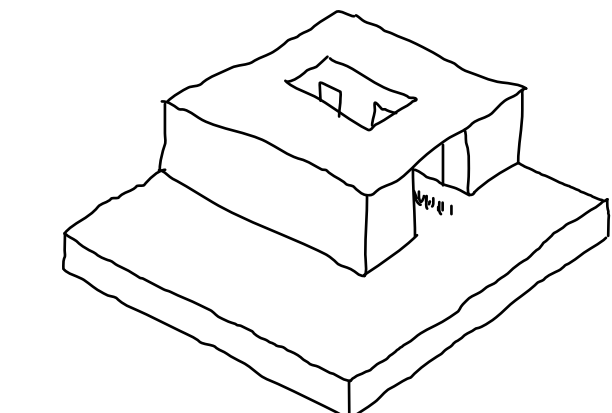
Rest



Courtyard



Dormitory



Pods

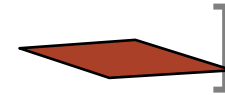
Residential zone

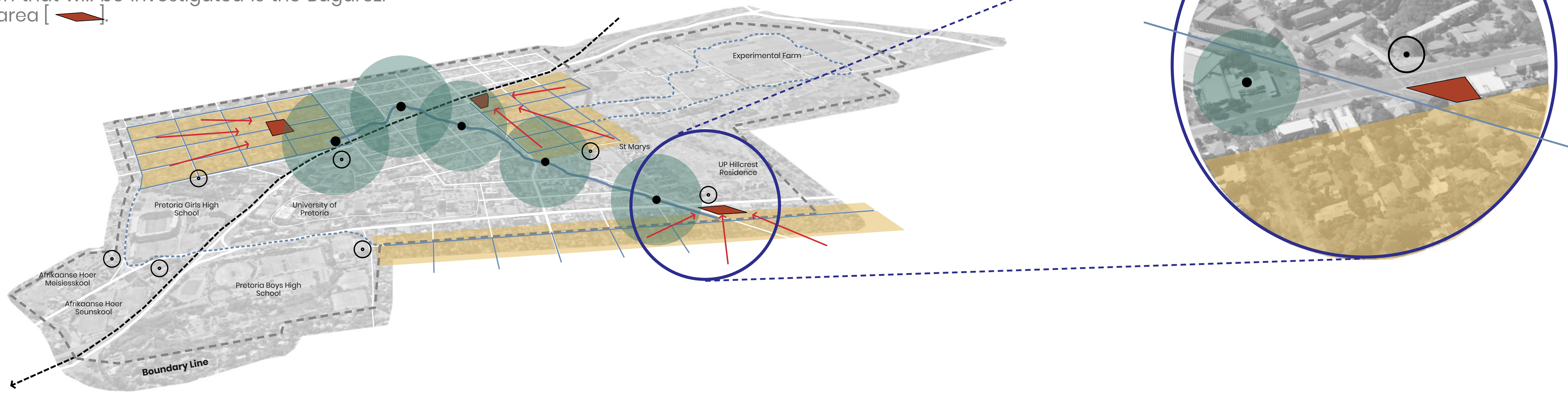
Garage zone

Stitching zone

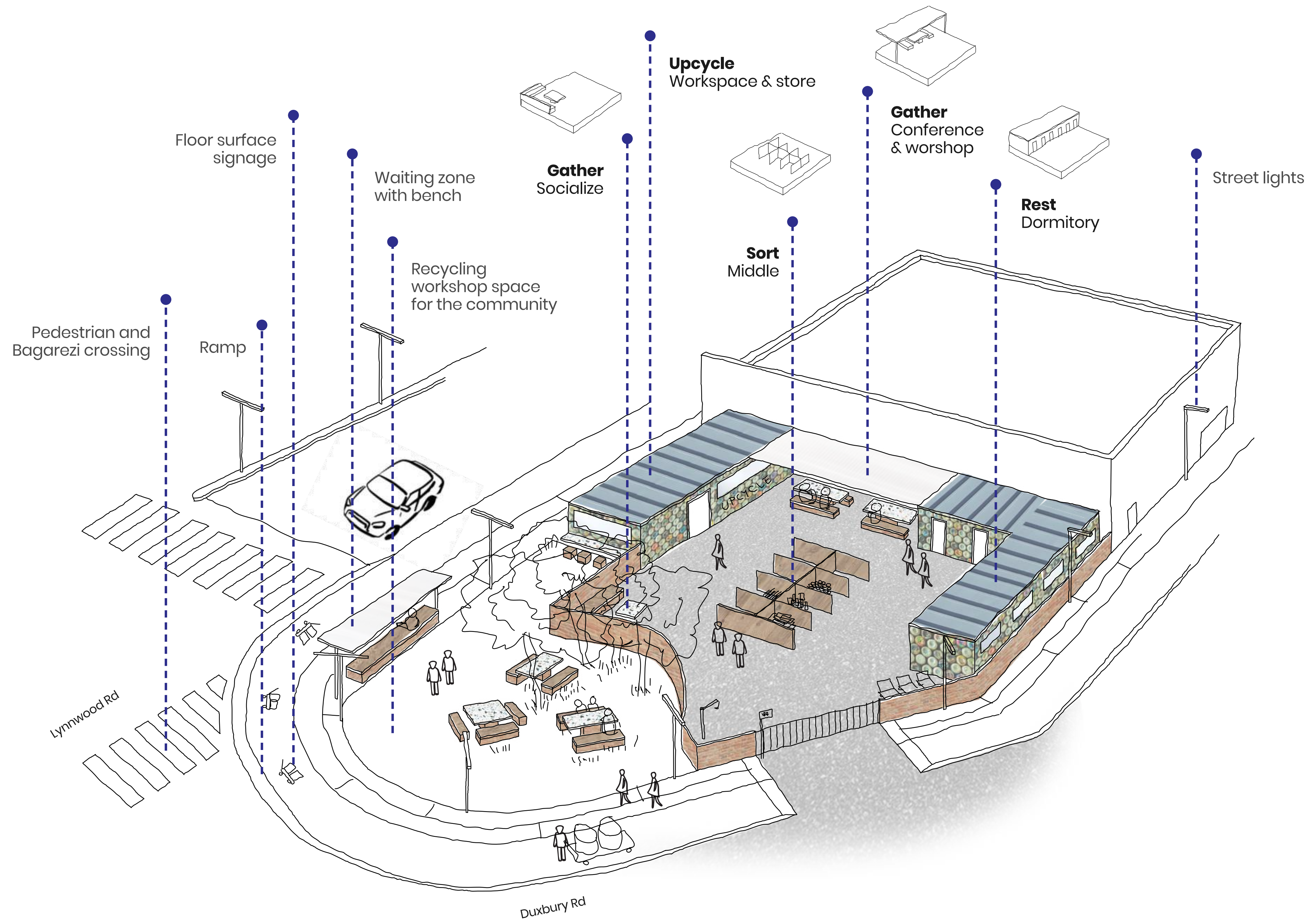
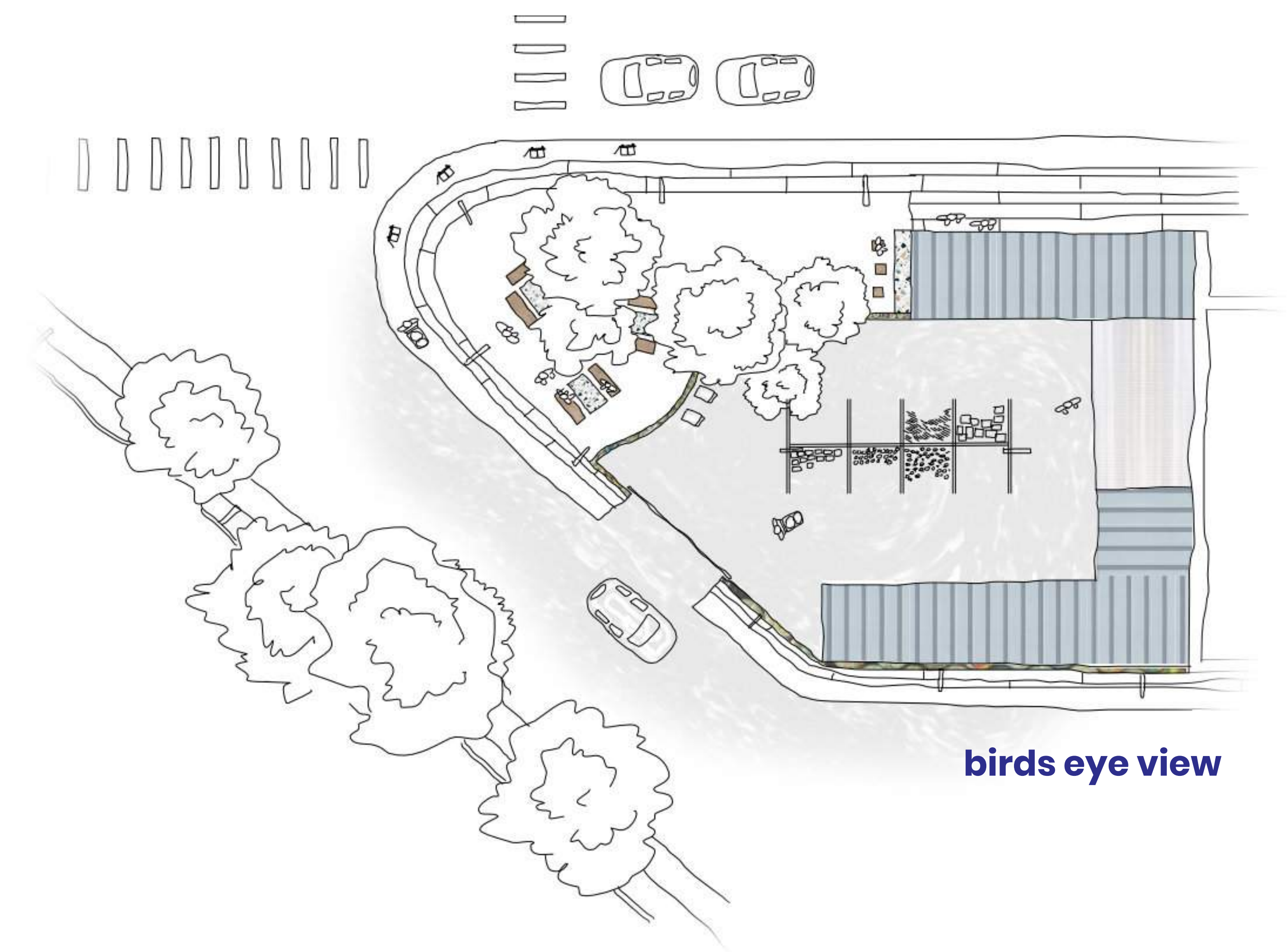
Bagarezi compound

Built Intervention

The first built intervention is located on the junction between Lynnwood and Duxbury rd, this intervention will act as a *benchmark* for all the other interventions to occur. The first consideration that will be investigated is the Bagarezi node within this area [].

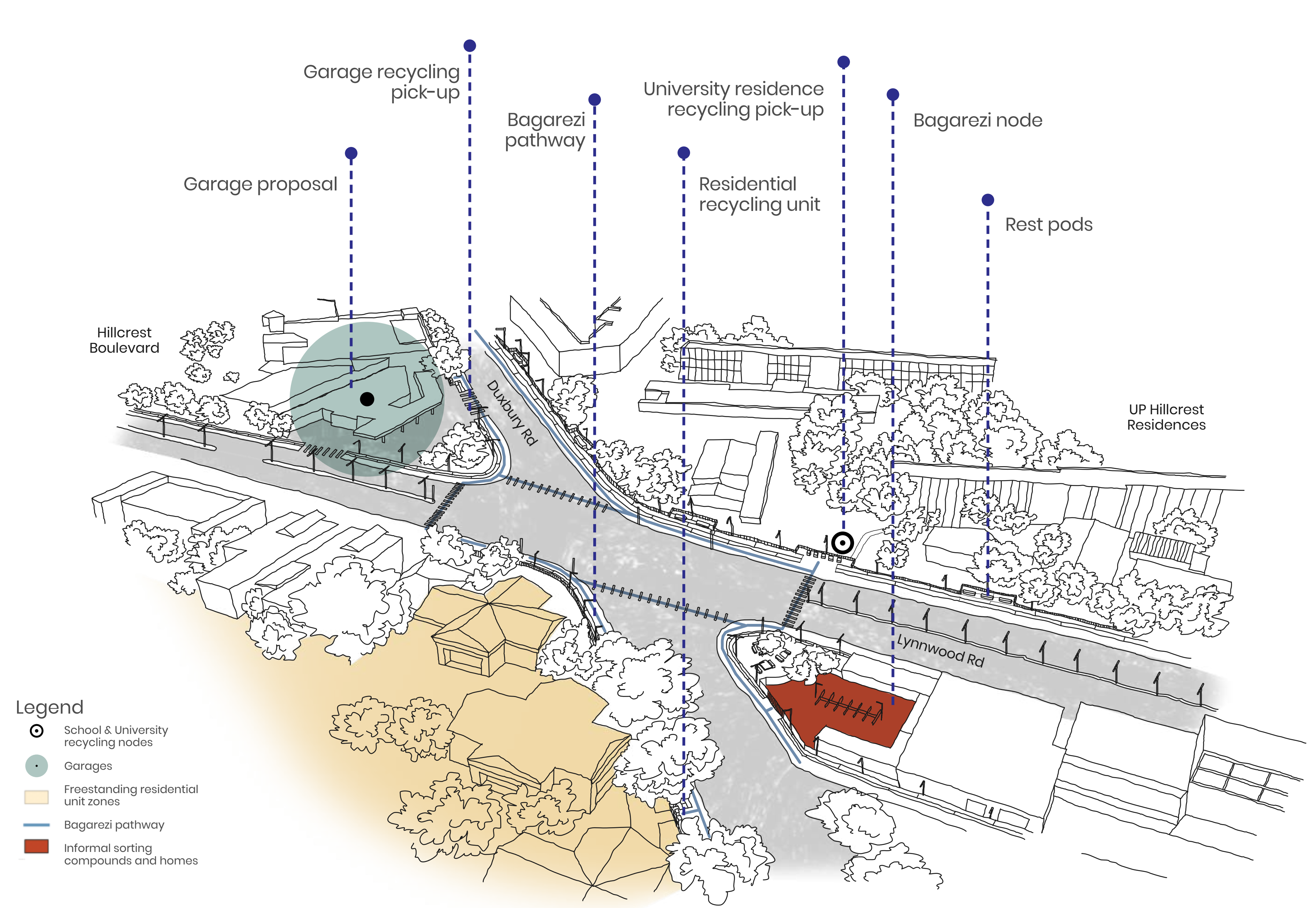


possible assembly of parts

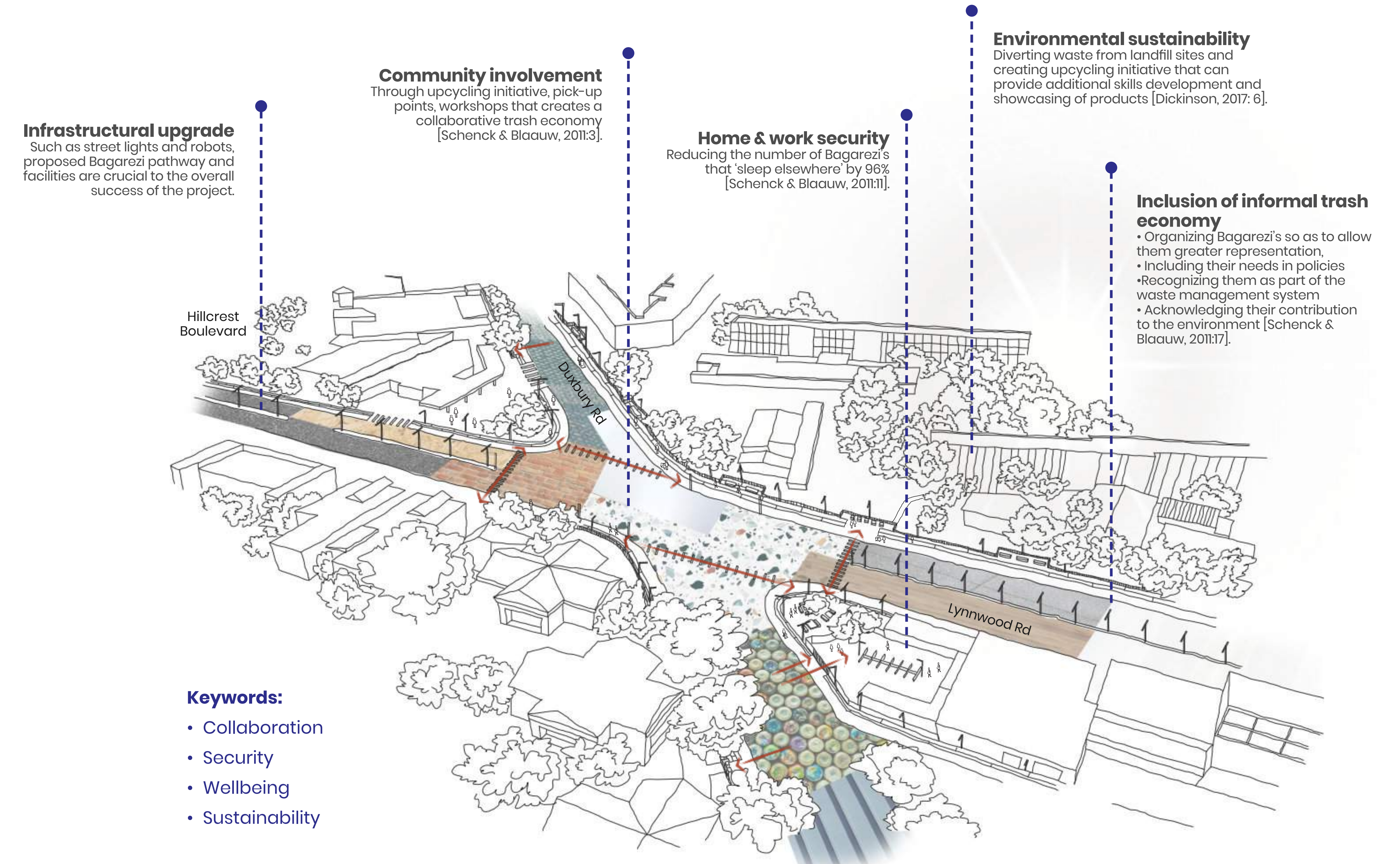


first phase outcome

built intervention



measurable outcomes



Reflection

challenges & opportunities

My approach to the project emphasized the importance of *deeply understanding the needs of waste pickers in order to design a system that would truly serve them*. Key challenges included *engaging with a marginalized community*, navigating the informal nature of their work, and addressing issues of home and work insecurity. To overcome these, I *reached out to relevant stakeholders and interacted directly with local waste pickers*, gaining crucial insights into their daily routines, struggles, and aspirations. This *user-centred approach* allowed me to inform design solutions that are *practical, meaningful, and responsive* to the people who will be most impacted.

accomplishments

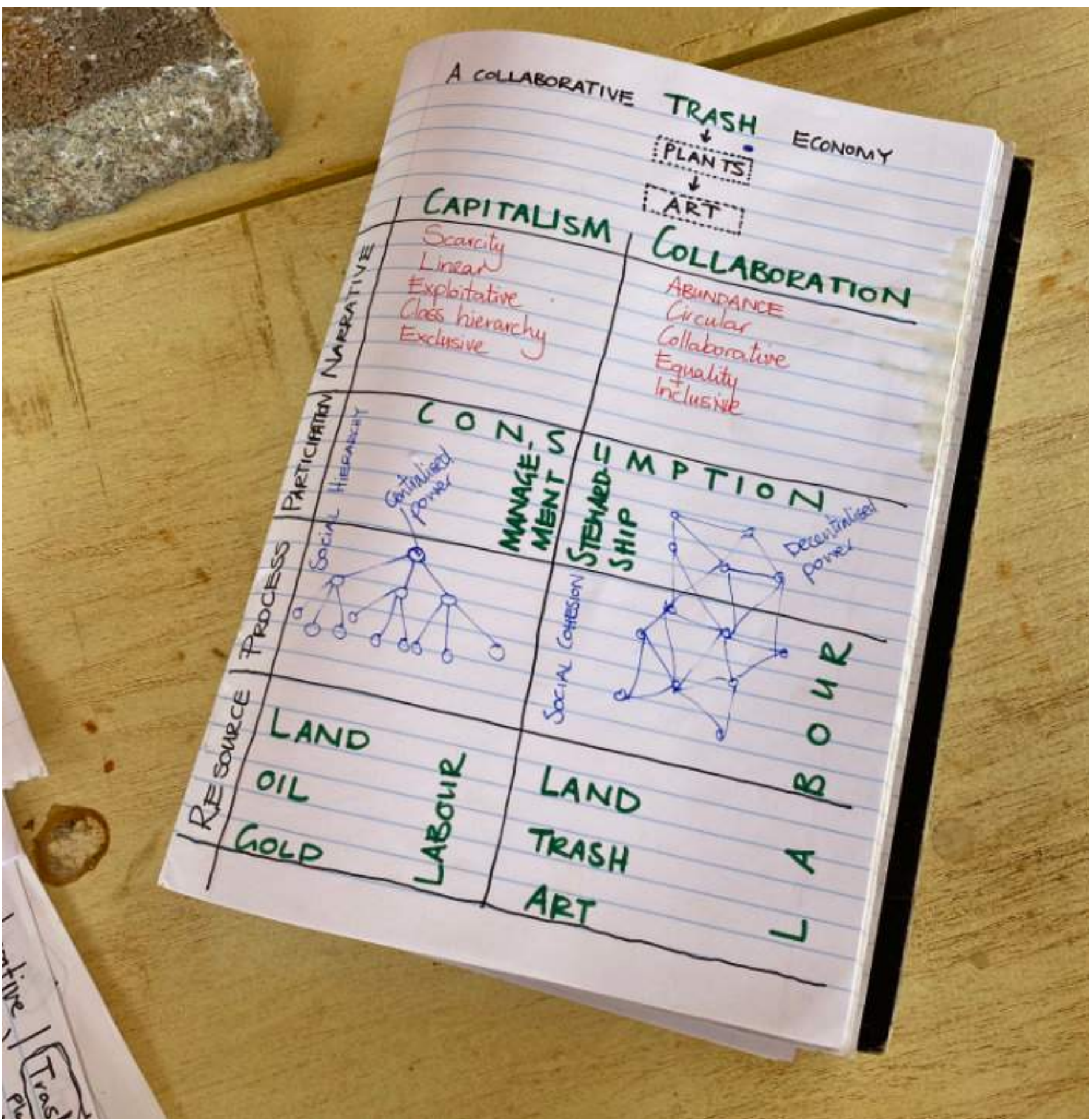
I achieved the highest grade in the class, earning a distinction for this project, which continued to serve as a studio precedent.

personal growth & insights

This project was profoundly moving, as it required *confronting the daily realities and hardships faced by the Bagarezis*. Engaging directly with their experiences highlighted the *human side of urban systems*, and it was humbling to discuss solutions while recognizing that meaningful change in the city takes time, effort, and persistence. The experience *strengthened my empathy and reinforced the importance of designing with compassion, and a long-term perspective*. It also deepened my appreciation for the impact *thoughtful, user-centred design* can have on vulnerable communities, and the need to *balance visionary ideas with practical, incremental steps toward social and infrastructural improvement*.



Vertical Studio is a week-long program where students from all years are grouped together to collaborate on a design that responds to the given brief.



Brainstorming the concept of a collaborative trash economy.



Informal sorting compound in Centurian.



Informal sorting compound in Centurian.



Informal sorting compound in Waterkloof.



I was given the opportunity to visit a number of sorting compounds in the surrounding areas of Hatfield by a non-profit organization [Project Zion] that is involved in their lives.



Urban farming in Hatfield [Moja Gabedi].